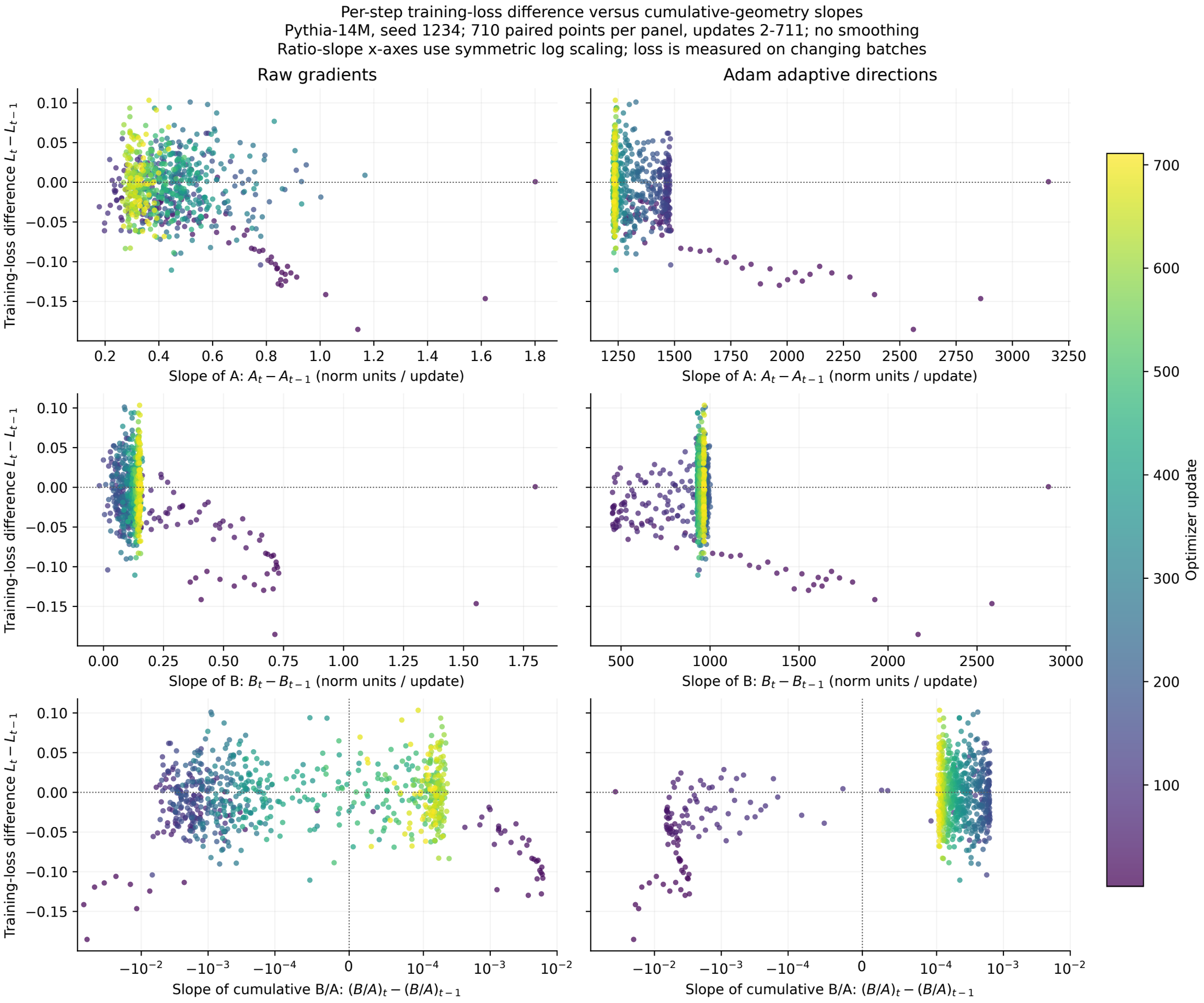

Per-step training-loss difference versus cumulative-geometry slopes
Pythia-14M, seed 1234; 710 paired points per panel, updates 2-711; no smoothing
Ratio-slope x-axes use symmetric log scaling; loss is measured on changing batches
Raw gradients
Adam adaptive directions
Training-loss difference $L_t - L_{t-1}$
Slope of A: $A_t - A_{t-1}$ (norm units / update)
Slope of B: $B_t - B_{t-1}$ (norm units / update)
Slope of cumulative B/A: $(B/A)_t - (B/A)_{t-1}$
Optimizer update